



GASKT: A Graph-Based Attentive Knowledge-Search Model for Knowledge Tracing

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Abstract. Knowledge tracking (KT) is a fundamental tool to customize personalized learning paths for students so that they can take charge of their own learning pace. The main task of KT is to model the learning state of the students, however the process is quite involved. First, due to the sparsity of real-world educational data, the previous KT models ignore the high-order information in question-skill; second, the long sequence of student interactions poses a demanding challenge for KT models when dealing with long-term dependencies, and the last, due to the complexity of the forgetting mechanism. To address these issues, in this paper, we propose a Graph-based Attentive Knowledge-Search Model for Knowledge Tracing (GASKT). The model divides problems and skills into two types of nodes, utilizing R-GCN to thoroughly incorporate the relevance of problem-skill through embedding propagation, which reduces the impact of sparse data. Besides, it employs the modified attention mechanism to address the long-term dependencies issue. For the attention weight score between questions, on the basis of using the scaled dot-product, the forgetting mechanism is fully considered. We conduct extensive experiments on several real-world benchmark datasets, and our GASKT outperforms the state-of-the-art KT models, with at least 1% AUC improvement.

Keywords: Knowledge tracing · Relational graph convolutional networks · Attention · LSTM · Forgetting mechanism

1 Introduction

Each student has his own pace of learning, but it is difficult to achieve personalized education in a traditional classroom. To address the distinct learning

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needs, the computer-aided education (CAE) system breaks fresh ground with its advanced technology. Knowledge tracing (KT), as a fundamental task in CAE systems, aims at evaluating the dynamic changes of students' knowledge states based on historical learning interactions.

There has been extensive research in the field of KT. However, the recent KT models mainly have the following three drawbacks. First, due to the sparseness of educational data, the high-order information between problems and skills has not been fully utilized. Second, because the student's interaction is a long sequence, it is generally appreciated that recurrent neural networks are not good at capturing such long-term dependencies. Finally, the complexity of the human brain's forgetting mechanism remains a major challenge.

Deep Knowledge Tracing (DKT) [1] is the first deep KT method, which exploits the Long Short-Term Memory (LSTM) [2] to predict the students' future performance. Since DKT uses one hidden variable to represent the knowledge state, it's difficult to trace the dynamic learning process of students. To address this limitation, Dynamic Key-Value Memory Networks (DKVMN) [3] extends on the memory-augmented neural networks (MANNs), using key matrix to store the representation of the skills, and the value matrix to characterize the knowledge state of each skill.

However, neither DKT nor DKVMN fully utilizes the high-order information in problem-skill, or adequately takes into account the forgetting behavior of students. Moreover, they are based on RNN mode and fail to effectively capture long-term dependencies between exercises.

DKT-forget [4] takes the memory mechanism into account based on DKT, adding three forgetting features, but still suffers from the remaining two shortcomings. Exercise-Enhanced Recurrent Neural Network (EERNN) [5] obtains embeddings of exercises from their textual information, aiming to obtain rich information to address the challenge of the scarcity of education data. However, in reality, textual information about the problem is difficult to obtain and EERNN fails to sufficiently consider the forgetting mechanism. Graph-based Interaction model for Knowledge Tracing (GIKT) [6] uses graph convolutional network (GCN) [7] to get the high-order embedding of the problems and skills, and introduces a history recap module to select the top-k most relevant problems by the attention score. However, in GIKT, insufficient consideration is given to the memory mechanism. For example, although question e_τ is close to the current question e_t , it will be filtered out due to its low score, which is inconsistent with the fact that recent information is clearer in short term memory.

To address the issues in current KT models, inspired by DKT-forget [4], EERNN [5] and GIKT [6], our Graph-based Attentive Knowledge-Search Model for Knowledge Tracing (GASKT) combines the strengths of the memory mechanism and high-order embedding. Specifically, unlike EERNN which uses bidirectional LSTM to encode textual information, we leverage the graph neural network to get the embedding of the exercises. Moreover, observing that the problems and skills belong to two different types of nodes, we transform the relationship between them into a heterogeneous graph, and utilize Relational Graph

Convolutional Network (R-GCN) [8] instead of GCN (in GIKT) to extract high-order information. Based on the use of LSTM to model the knowledge state of students, the attention mechanism is applied to further enhance the model's ability of capturing long-term dependencies, which is thoroughly incorporated with forgetting behavior.

In summary, in this paper, we propose a deep KT framework GASKT. Our main contributions are as follows:

1. We use R-GCN to extract high-order information about problems and skills, and divide problems and skills into two types of nodes, then construct a corresponding heterogeneous graph.
2. We give adequate consideration to the forgetting mechanism and employ the modified attention mechanism to evaluate students' future performance. Based on the similarity computed by the scaled dot-product, we combine two more features into the attention weight scores between questions: time distance and past trial counts.
3. We conduct experiments on three benchmark datasets, and the results show that GASKT has better performance and higher interpretability compared with existing solutions.

2 Related Work

Knowledge tracing (KT) is the task of capturing the dynamic changes of students' knowledge state based on their historical learning interactions.

The existing knowledge tracing methods can be roughly categorized into three main types, Bayesian knowledge tracing (BKT), factor analysis models, and deep learning models.

BKT. [9] models the student's knowledge state as a binary-valued variable, and uses the hidden Markov model to update and maintain the variable based on the student's practice. Several studies have integrated some other factors, such as individual student's prior knowledge [10], slip and guess probabilities [11] and item difficulty [12].

Factor Analysis. Among the factor analysis models, The additive factor model (AFM) [13] considers the number of students' attempts to the question, while the performance factor analysis model (PFA) [14] counts the positive and negative attempts separately.

Deep Learning Methods. In deep KT models, Deep Knowledge Tracing (DKT) [1] is the first model that applies deep learning to KT. Another notable model is Dynamic Key-Value Memory Networks (DKVMN) [3], which extends on memory-augmented neural networks (MANNs), using key matrix to store the representation of skills and value matrix to represent and update students' knowledge acquisition ability. DKT-forget [4] builds on DKT by incorporating three forgetting properties in the input of the model. And DKT+ [15] is an

improved version of DKT that takes into account the regularization of prediction consistency.

However, both DKT and DKVMN are based on RNN [16], these models and their variants suffer from impoverished capturing capabilities when dealing with long sequences. To tackle this issue, the attention mechanism has been extensively used in many studies. Self Attentive Knowledge Tracing (SAKT) [17] is the first model to incorporate the self-attention mechanism in the KT. Given that people’s memory declines exponentially over time, and repetitive practices deepen the user’s memory, it’s clear that the plain attention mechanism fails to satisfy the requirements of KT, so our GASKT modifies the attention mechanism by combining the forgetting features of the learner: the time distance and the past trial counts.

To cope with the challenges of sparse data, some models incorporate the textual content of exercises, such as Exercise-Enhanced Recurrent Neural Network (EERNN) [5] extracts textual features via a bidirectional LSTM. Another way to resolve this issue is to leverage graph neural networks. Graph-based Interaction Model for Knowledge Tracing (GIKT) [6] is the first KT model that uses GCN to learn high-order information in problem-skill. Considering that it is difficult to obtain the textual information of the exercise in reality, we utilize graph neural networks to learn the embedding of the nodes. Different from the GIKT, our proposed GASKT employs R-GCN rather than GCN to extract high-order information between nodes, by constructing the problem-skill relationship graph into a heterogeneous graph.

3 Problem Definition

The problem of knowledge tracing (KT) can be described as follows: when a student is answering the question e_t , given his historical learning record $X = (x_1, x_2, \dots, x_{t-1})$, where $x_i = (e_i, r_i)$, $r_i \in \{0, 1\}$ is the correctness of the student’s answer to the question e_i . The goal is to predict the probability that the student will answer the current question e_t correctly. Particularly, in our work, we denote the total length of questions as $|E|$, the total number of skills as $|S|$, and the number of questions answered by each student is unified as l .

Question embedding matrix: $\tilde{\mathbf{E}} \in \mathbb{R}^{|E| \times d}$, where d denotes the embedding dimension. For a problem e_i its node embedding is $\tilde{e}_i$, corresponding to row i in the matrix $\tilde{\mathbf{E}}$. Similarly, $\tilde{\mathbf{E}} \in \mathbb{R}^{l \times d}$ represents the embedding matrix of the questions answered by the student.

Skill embedding matrix: $\tilde{\mathbf{S}} \in \mathbb{R}^{|S| \times d}$, where d is the same embedding dimension as question embedding matrix. For a skill s_i its node embedding is $\tilde{s}_i$, corresponding to row i in the matrix $\tilde{\mathbf{S}}$.

Student’s knowledge states matrix: $\mathbf{H} \in \mathbb{R}^{l \times d_h}$, where d_h represents the embedding dimension of the student’s knowledge state. Here, row i of the $\mathbf{H}$

matrix represents the embedding representation of the student’s knowledge state at timestamp i .

Student’s attention scores matrix: $\mathbf{A} \in \mathbb{R}^{l \times d_h}$. In the $\mathbf{A}$ matrix, row i represents the embedding representation of the student’s attention score at timestamp i .

4 The GASKT Method

This section will clarify our framework in detail. GASKT comprises four components, embedding module, knowledge evolution module, knowledge search module, and the response prediction module. Figure 1 visualizes the overall framework of the model.

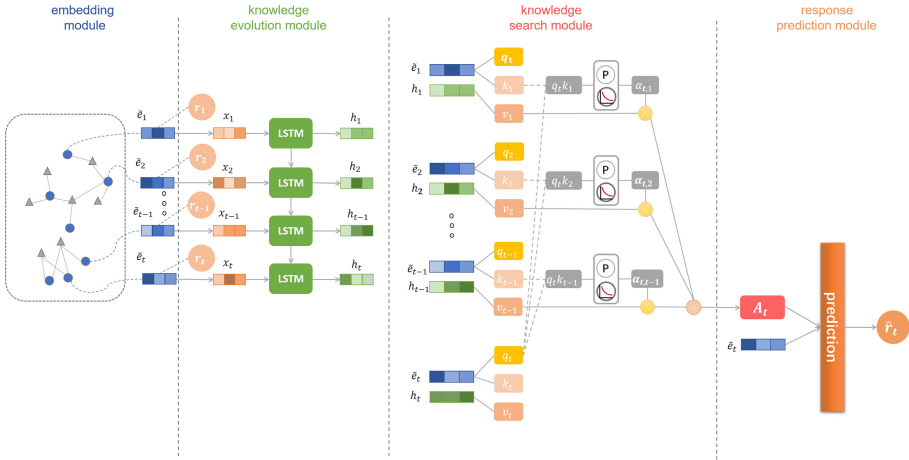


Fig. 1. The overall framework and workflow of GASKT

4.1 Embedding Module

The correctness of students’ responses to the new question predominantly depends on their mastery states of the similar questions and skills. Consequently, the feature embedding between questions and skills is particularly important. However, the sparsity of available educational data brings a demanding challenge to the predictions for KT. In this model, to meet the challenge, we use the R-GCN [8] to fully extract question-skill correlations.

R-GCN [8] is an extension to GCN [7] that aims at handling large-scale multi-relational data. The key difference between them is that GCN contains the same type of nodes and edges, so they share the same weight parameters; while R-GCN has different types of nodes and edges, they will share the same parameters only if they belong to the same type. Obviously, the problems and

skills belong to different types of nodes, so we choose R-GCN to represent the relationship map among them in the form of a heterogeneous graph.

We construct a directed and labeled multi-graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{R})$, where node $v_i \in \mathcal{V}$, labeled edge (relation) $(v_i, r, v_j) \in \mathcal{E}$, $r \in \mathcal{R}$ is the relationship type. In this model, there are two types of nodes, problem nodes e and skill nodes s ; two types of edges, that is, $(problem, contain, skill)$, $(skill, contain - by, problem)$.

The calculation of R-GCN consists of the following two steps. First, it calculates and aggregates the messages for each relation r , and second, it aggregates the results from multiple relations. The R-GCN formula for the l level of node v_i can be expressed as follows:

$$h_i^l = \sigma\left(\sum_{r \in \mathcal{R}} \sum_{j \in \mathcal{N}_i^r} \frac{1}{c_{i,r}} W_r^{(l-1)} h_j^{(l-1)} + W_o^{(l-1)} h_i^{(l-1)}\right) \quad (1)$$

where $\mathcal{N}_i^r$ denotes all neighbor nodes of node v_i under relation $r \in \mathcal{R}$. W_r, W_o is the weight parameter of the network, $c_{i,r}$ is a normalization constant, and σ is a nonlinear transformation.

4.2 Knowledge Evolution Module

We concatenate the student's answer r_t with the embedding feature $\tilde{e}_t$ of exercise e_t , besides, the r_t is extended to $\mathbf{r}_t = [r_t, r_t, \dots, r_t] \in \mathbb{R}^d$.

$$x_t = [\tilde{e}_t, \mathbf{r}_t] \quad (2)$$

The sequence dependencies between exercises in the interactions will severely affect the learning state of students. In order to capture these dependencies, we use a Long Short-Term Memory Network (LSTM) [2] to model the learning process of students.

$$i_t = \sigma(W_i[x_t, h_{t-1}, c_{t-1}] + b_i) \quad (3)$$

$$f_t = \sigma(W_f[x_t, h_{t-1}, c_{t-1}] + b_f) \quad (4)$$

$$o_t = \sigma(W_o[x_t, h_{t-1}, c_{t-1}] + b_o) \quad (5)$$

$$c_t = f_t c_{t-1} + i_t \tanh(W_c[x_t, h_{t-1}] + b_c) \quad (6)$$

$$h_t = o_t \tanh(c_t) \quad (7)$$

Where i_t , f_t , o_t , c_t , and h_t represents input gate, forget gate, output gate, cell state, and hidden state respectively. Finally, we get the student's knowledge state matrix $\mathbf{H} \in \mathbb{R}^{|I| \times d_h}$.

4.3 Knowledge Search Module

In the knowledge evolution module, we use LSTM [2] to model the student's knowledge state. However the LSTM has limited capacity to capture long-term dependencies when dealing with long sequences. To tackle this limitation, we

apply the modified attention mechanism to enhance the model's ability to capture information.

Same as the basic scaled dot-product attention mechanism, we need three inputs as Q , K and V . In this model, we map the embedding matrix $\tilde{\mathbf{E}} \in \mathbb{R}^{l \times d}$ of the questions answered by the student to Q , K respectively (the dimensions of them are d_q, d_k , where $d_q = d_k$), and map the student's knowledge mastery state $\mathbf{H} \in \mathbb{R}^{l \times d_h}$ to V (the dimension is d_v).

With the basic scaled dot-product attention mechanism, we compute the attention distribution as follows:

$$\alpha_{t,\tau} = \text{Softmax}\left(\frac{Q_t K_\tau^T}{\sqrt{d_k}}\right) = \frac{\exp\left(\frac{Q_t K_\tau^T}{\sqrt{d_k}}\right)}{\sum_{\tau'} \exp\left(\frac{Q_t K_{\tau'}^T}{\sqrt{d_k}}\right)} \in [0, 1] \quad (8)$$

As we know the current state of knowledge is obtained based on the student's historical performance. Therefore, when calculating the attention score, for the question e_t , we only consider questions e_τ that satisfies $\tau < t$.

Intuitively, Q_t indicates the mapping of $\tilde{e}_t$, which is the embedding representation of the question e_t answered at time t , and K_τ indicates the mapping of $\tilde{e}_\tau$. Where $\alpha_{t,\tau}$ represents the effect of question e_τ on question e_t , that is, the similarity between them. Exactly, the closer the $\alpha_{t,\tau}$ is to 1, the greater the similarity between the two questions.

H. Ebbinghaus [18] has mentioned in his early research that people's memory declines exponentially over time, and repeated practice can deepen users' memory. Given the forgetting mechanism of the human brain, the basic attention mechanism fails to satisfy the requirements of KT. Therefore, combined with [4, 18], we add two forgetting factors to the basic attention formula, one is the time distance, and the other is the past trial counts.

- **time distance**: time distance between two arbitrary problems.
- **past trial counts**: the number of questions that student has previously answered with the same skills as the current one.

According to the human brain's forgetting mechanism, the influence between e_t and e_τ tends to decrease exponentially with time. Additionally, considering that repetition deepens the human brain's memory, the influence between them should be greater when problems shared with the same skills as e_τ have been trained repeatedly. Combining the above memory mechanisms, we redefine the attention formula as follows:

$$\alpha_{t,\tau} = \text{Softmax}\left(\frac{e^{-\theta D_{t-\tau}} \cdot e^{\beta P_\tau} \cdot Q_t K_\tau^T}{\sqrt{d_k}}\right) \in [0, 1] \quad (9)$$

Where $D_{t-\tau}$ is $D_t - D_\tau$, which denotes the distance between problem e_t and problem e_τ . θ, β are decay factors. $e^{-\theta D_{t-\tau}}$ denotes the factor of the distance. P_τ represents the total number of practiced exercises that share the same skill as question e_τ before time τ . And $e^{\beta P_\tau}$ denotes the factor of the past trial counts.

Moreover, we incorporate the multi-head attention mechanism to jointly focus on different parts of the input information, allowing the model to summarize the learner’s past learning performance in multiple time dimensions. More specifically, we use h independent attention heads (where $d_q = d_k = d/h, d_v = d_h/h$), and concatenate the final output into a $1 \times (h \cdot d_v)$ vector, where $h \cdot d_v = d_h$.

The final attention score matrix for all problems that answered by a student is as follows.

$$\mathbf{A} = \text{Attention}(Q, K, V) = \sum_t \sum_{\tau} \alpha_{t,\tau} v_{\tau} \quad (10)$$

Where $\mathbf{A} \in \mathbb{R}^{l \times d_h}$, represents the student’s attention scores matrix.

4.4 Response Prediction Module

The response prediction module is a fully connected network with sigmoid function. To predict the probability that a student will answer question e_t correctly, we concatenate A_t (student’s attention score of the current question e_t) with $\tilde{e}_t$ (embedding of the current question e_t) as the input to the response prediction module.

$$\hat{r}_t = \sigma(W_r[A_t, \tilde{e}_t] + b_r) \quad (11)$$

The parameters of the entire model GASKT can be optimized by minimizing the cross-entropy loss $\mathcal{L}$ between the predicted probability of the correctness $\hat{r}_t$ and the true label of the student’s answer r_t .

$$\mathcal{L} = - \sum_t (r_t \log \hat{r}_t + (1 - r_t) \log(1 - \hat{r}_t)) \quad (12)$$

5 Experimental

5.1 Datasets

Table 1. Datasets

	ASSIST09	ASSISTChall	Statics
Users	3841	1708	333
Ques	15911	2210	-
Skills	123	101	1223
Inter	258896	864711	189297

Table 2. Performance comparison

Model	ASSIST09	ASSISTChall	Statics
BKT	0.6571	0.7301	0.7300
DKT	0.8024	0.7263	0.8233
DKT+	0.8236	0.7339	0.8175
DKVMN	0.8093	0.7073	0.8195
GIKT	0.8108	0.7586	0.8230
GASKT	0.8301	0.7800	0.8271

To verify the effectiveness of the GASKT model, we conduct a series of experiments on three benchmark datasets. Table 1 illustrates the statistics of the datasets.

- **ASSIST09**¹: This dataset was collected by ASSISTment online tutoring platform in 2009-2010. Refer to the standard processing of the previous work [19], we delete users with less than 3 interactions and remove the problem of empty skills and scaffolding, besides.
- **ASSISTChall**²: This dataset was obtained from the 2017 data mining competition. The data processing method is the same as ASSIST09.
- **Statics**³: This dataset was collected from a one-semester Statics course through an online educational system. We process the dataset the same as ASSIST09.

5.2 Baseline

To validate the performance of the model, we select the following models as the baseline.

- **BKT** [9] models the student’s knowledge state as a set of binary variables on the basis of Bayesian theory.
- **DKT** [1] is the first model that applies deep learning methods to KT. It uses LSTM to model the knowledge states of students.
- **DKT+** [15] adds the regularization of prediction consistency on the basis of DKT.
- **DKVMN** [3] extends on memory augmentation networks (MANNs) by using a key matrix to store the concepts of knowledge and a value matrix to update the student’s knowledge state.
- **GIKT** [6] aggregates problems and skills embeddings by using graph convolutional networks.

5.3 Implementation Details

To set up the experimental conditions, we randomly split the dataset into 3:1:1, with 60% for training, 20% for validation, and 20% for testing.

To improve computational ability, we refer to previous work [3] and uniformly set the length of student interaction to 200. For the multi-skilled questions (ASSIST09) [1,3], we present the questions multiple times, each corresponding to a skill. In the embedding module, the dimension of the embedding size d is set to 128, and the maximum number of the sampling layers L is set to 3. Next in the knowledge evolution module, set the output dimension d_h of LSTM to 128, representing the size of student’s knowledge state dimension.

To prevent overfitting, we used a dropout with probability 0.2. In addition, Adam optimizer is utilized to find the optimal parameters of the model, with the learning rate at 0.001 and the batch size of 24. Except for the Statics, since it is small, we set its batch size to 12.

¹ <https://sites.google.com/site/assistmentsdata/home/assistment-2009-2010-data/skill-builder-data-2009-2010>.

² <https://sites.google.com/view/assistmentsdatamining>.

³ <https://pslcdatashop.web.cmu.edu/DatasetInfo?datasetId=507>.

5.4 Overall Performance

Table 2 records the AUC performances of all KT models involved in the comparison. As can be seen from the table, our proposed GASKT achieves the best performance for predicting students' future knowledge state over three benchmark datasets, which verifies the effectiveness of GASKT.

In conclusion, we have the following observations. First, all deep KT models perform better than BKT. Second, the latest deep KT methods show greater improvements than DKT, suggesting that it is insufficient to represent the knowledge state with only one hidden variable. Third, our proposed GASKT achieves an average performance improvement of 1.90% compared to GIKT. In our experimental setting, multi-skilled problems are repeated multiple times, with each corresponding to a skill. However, in GIKT, joint skills are used as skill representation of the problem. To standardize the criteria, the same experimental setup as ours is used for GIKT, and we observe that the AUC performance of GIKT on the ASSIST09 is 0.8108, slightly better than its work report of 0.7896. On ASSIST09 and ASSISTChall, our GASKT significantly outperforms GIKT, which illustrates that taking forgetting mechanisms into account is more conducive to simulating the knowledge states of students. More importantly, on Statics, since it only contains skills, both GASKT and GIKT perform random assignments to the nodes of this dataset. In this setting, our GASKT has a slight performance improvement over GIKT, which shows that in small datasets, the effect of the modified attention mechanism is not significant compared with large data sets.

5.5 Ablation Studies

Effect of R-GCN Layer. Table 3 lists the performance of GASKT with changes of L . In particular, when $L = 0$, the node embedding is initialized randomly.

Table 3. Effect of R-GCN layer

	ASSIST09	ASSISTChall
0	0.8125	0.7760
1	0.8291	0.7768
2	0.8294	0.7783
3	0.8301	0.7800

Table 4. Effect of the modified attention

Model	ASSIST09	ASSISTChall	Statics
GASKT-O	0.8171	0.7686	0.8197
GASKT-D	0.8286	0.7789	0.8263
GASKT-N	0.8187	0.7720	0.8206
GASKT	0.8301	0.7800	0.8271

The experimental results show that with an increase of L , the performance of GASKT gradually improves over the datasets. When $L = 3$, the model achieves the highest AUC performance, which proves the effectiveness of high-order aggregation to the model. Notably, on the ASSIST09 dataset, the greatest improvement in model performance is seen with successive increases of L . More specifically, when $L = 0$, the AUC performance of GASKT is 0.8125, and when $L = 3$,

the performance is 0.8301, with a significant improvement of 2.17%. However, the model performance improves marginally by 0.52% on the ASSISTChall. Based on phenomenon mentioned above, it can be inferred that it is because ASSIST09 contains the most number of problems and skills in these datasets and has multi-skilled problems, which allows the graph neural network to outperform the random initialization in learning node embedding.

Effect of the Modified Attention. To verify the effectiveness of the two forgetting factors, we perform the following experiments. The detailed results of the experiment can be found in Table 4.

- **GASKT-O** Based on the original attention formula, which fails to consider the forgetting mechanism.
- **GASKT-D** The effect of the time distance is added to the original attention function.
- **GASKT-N** The influence of the past trial counts is added to the original attention function.
- **GASKT** Taking the forgetting mechanism into consideration, adding two elements: time distance and the past trial counts.

From the table, it can be found that GASKT-D and GASKT-N show greater improvement over GASKT-O, verifying that the forgetting factors facilitate the model performance, which is consistent with the pattern of memory mechanisms. Furthermore, taking GASKT-O as reference, the AUC performance of GASKT-D has considerably improved by an average of 1.18% over three datasets, while the performance of GASKT-N has slightly increased by an average of 0.25%, signifying that time distance has a greater impact on the current knowledge state compared to the past trial counts.

6 Conclusion

In this paper, we proposed the GASKT model, which focused on predicting the learner’s mastery of the current exercise. Our method extracted high-order information between questions and skills via R-GCN, and combined the attention mechanism with the LSTM to deeply explore the dependencies between the current exercise and the previous exercises. Furthermore, to take forgetting mechanisms into account, we incorporated two forgetting factors, time distance and past trial counts. The experimental results demonstrated that our method outperformed the most previous KT models on three benchmark datasets, and exhibited outstanding interpretability.

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